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# Systems and Methods for Big and Unstructured Data Project

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# 1 | Introduction

## 1.1. Problem Specification

The aim of this project was to design a 'query graph data structure' in Neo4j for supporting a bibliographic Database. The database must store a large scale Dataset recording publications with their features and their descriptions in order to have an overview of the library.

## 1.2. Hypothesis

The assumptions taken into account are the following:

- All data are assumed to be truthful without further verification;
- We are using the concept of "publication" referring to Book, Article, Thesis, Online Resource, Proceeding, Inproceeding and Incollection, instead, Journal isn't belonging to this family of Entities;
- Whoever uploads the data relating to publications does so by entering all the information in a correct and complete way;
- An author doesn't publish the same work more than once so that redundancy and duplicates are avoided in the system;
- Every publication has an associated dynamic attribute "CitationScore" that computes the number of works that quote a specific one. Every author has his own "CitationScore" attribute too, it is computed with a formula which aims to balance the score between the authors. As said, this attribute is dynamic and any change due to references updating can be managed through different approaches:
    1. The attribute can be kept inconsistent in the db until the next reading when it will be recalculated. This is the worst case scenario because the number of reading is much higher than the writing one,therefore this behavior is inefficient;

2. The attribute can be kept inconsistent in the db until it will be recalculated by a trigger after a certain amount of time. This context is not entirely problem-free because there are small chances that the trigger doesn't update the value in time.

3. The attribute is always kept consistent after an applied operation. The score is update after three precise operation about the publication: create,delete and update. All this is fine because the writing operation frequency is lower than the reading one.

- Cite is a randomly generated relationship between publications, the references don't quote works outside the local db.
- The editor of the publication is consider as the author.

# 2 | ER Diagram

## 2.1. Entities and Relationships

The designed ER diagram contains the following entities: Person generic, Researcher, Book, Article, Journal, Proceeding, Inproceeding, Incollection, Publisher, Thesis, Online Resource. Some classes have been introduced due to ER diagram correctness reasons, but they haven't been taken into account during the design of the Graph Database for ease of management of the DB as delign in 3.2. Entities are retrieved through a process of data analysis in which the hierarchy has been structured according to the XML file, instead, relationships and attributes are selected in a way that only the most relevant ones are kept for the project.

## 2.2. ER Diagram description

The Person Generic Entity is a Superclass identified by the CF and its specialized subclass is Researcher with a unique attribute Orcid which provides a persistent digital identifier distinguishing every researcher, this choice has been made to differentiate an amateur that doesn't make any formal publication from a professional researcher. The generalization is disjoint and partial. Every publication such as Book, Article, Thesis, Online Resource, Proceeding, Inproceeding, Incollection is made unique by its key and its publication date. Proceeding is a collection of academic papers published in the context of an academic conference or workshop which contains the contributions made by researchers(Inpreceeding) at the conference. The Entity Incollection is referred to the entries that are written from an author that is different from the one that has drawn up and edited the book. Volume, Number and Series, as said before, have been introduced due to correctness aspect but they are represented only as relationship in the graph diagram. Every Journal has its name, Volume is identified by Journal and Volume Number, Number is identified by Volume and Number. Series are collections of publications which refer to a common topic or theme, they are identified by their title. These are the attributes contained by each relationship:

- Book_in between Book and Incollection: Page end and Page start;
- incollection_Series: Number;
- Book_Series: Number;
- Preceeding_Series: Number;
- Article_Number: Page end and Page start.

The ER model is attached in PDF

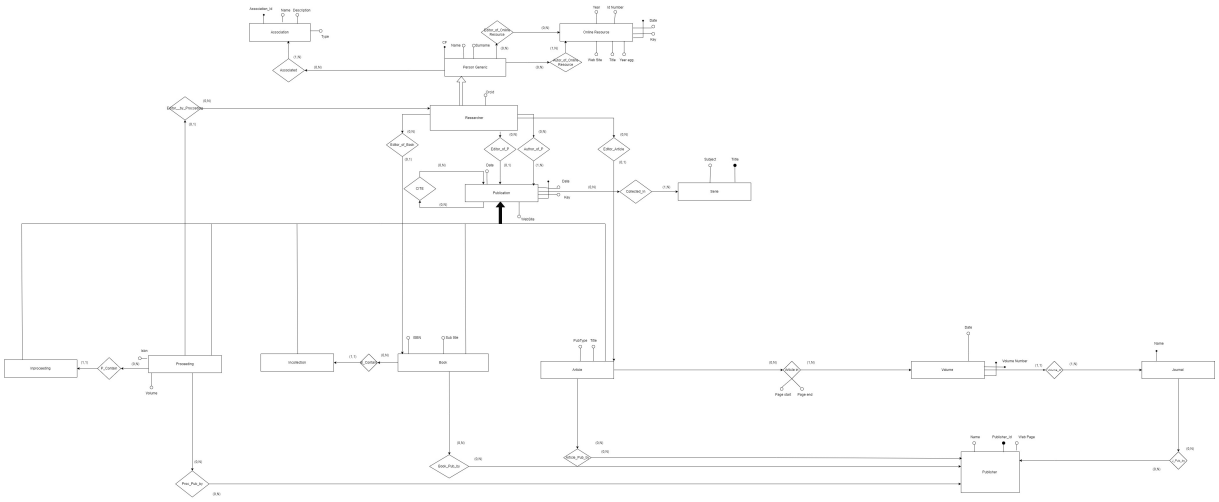


Figure 2.1: ER Diagram

# 3 | Graph Diagram

## 3.1. Graph Diagram description

Author, Article, Journal, Book, Incollection, Proceedings, Inproceedings, Www, Mastersthesis, University and Series are represented as nodes and their attributes are shown in the graph diagram below. The relationships of the database are WRITTEN_BY, CONTAIN, PUBLICATED_BY, COAUTHOR, AT, CITE, INCLUDE. We omitted some links using the publication note make graph diagram more readable and understandable.The nodes Article, Book, Incollection, Proceedings, Inproceedings, Www, Mastersthesis contain the Publication tag in order to perform and understand queries easily since you don't need to specify 7 different nodes in the MATCH statement every time. The relationships between nodes are the following:

```
(Pubblication)-[WRITTEN_BY]->(Author)
(Proceedings)-[CONTAIN]->(Inproceedings)
(Book)-[CONTAIN]->(Incollection)
(Mastersthesis)-[AT]->(University)
(Author)-[PAST_STUDENT]->(University)
(a1:Author)-[COAUTHOR]-(a2:Author)
(p1:Publication)-[CITE]->(p2:Publication)
(Article)-[PUBLICATED_BY]->(Journal)
(Serie)-[INCLUDE]->(Publication)
```

Publication and Author nodes have the citationScore attribute. In a particular for publication it is used to store a computed index which describes how many times that node is quoted. This attribute is updated by using a specific command that executes the PageRank algorithm. The citationScore attribute will be used in queries to classify authors and publications. The relationship COAUTHOR is introduced to increase the speed of the queries searching for the relationships between authors. Two authors are connected by the coauthor relationship only if they have participated to the writing of the same publication. The implementation of the relationship PAST_STUDENT was made to optimize

the queries on the MasterThesis publication type. Its purpose is to associate an author who wrote a thesis directly with its university. Journals are represented as nodes and described only by their name, it has a relationship with Article sharing the publication details (year, month, volume, pages, number). By doing this it is possible to efficiently aggregate all the articles published in the same journal without creating one node for each number of that journal. The database design decision follows two thoughts:

1. be fast, the core of the structure is able to easily guarantee the functioning of complex algorithm like shortest path, thanks to the quantity of relations;
2. be used for ranking authors and articles based on citations(for example we are refering to citationScore).

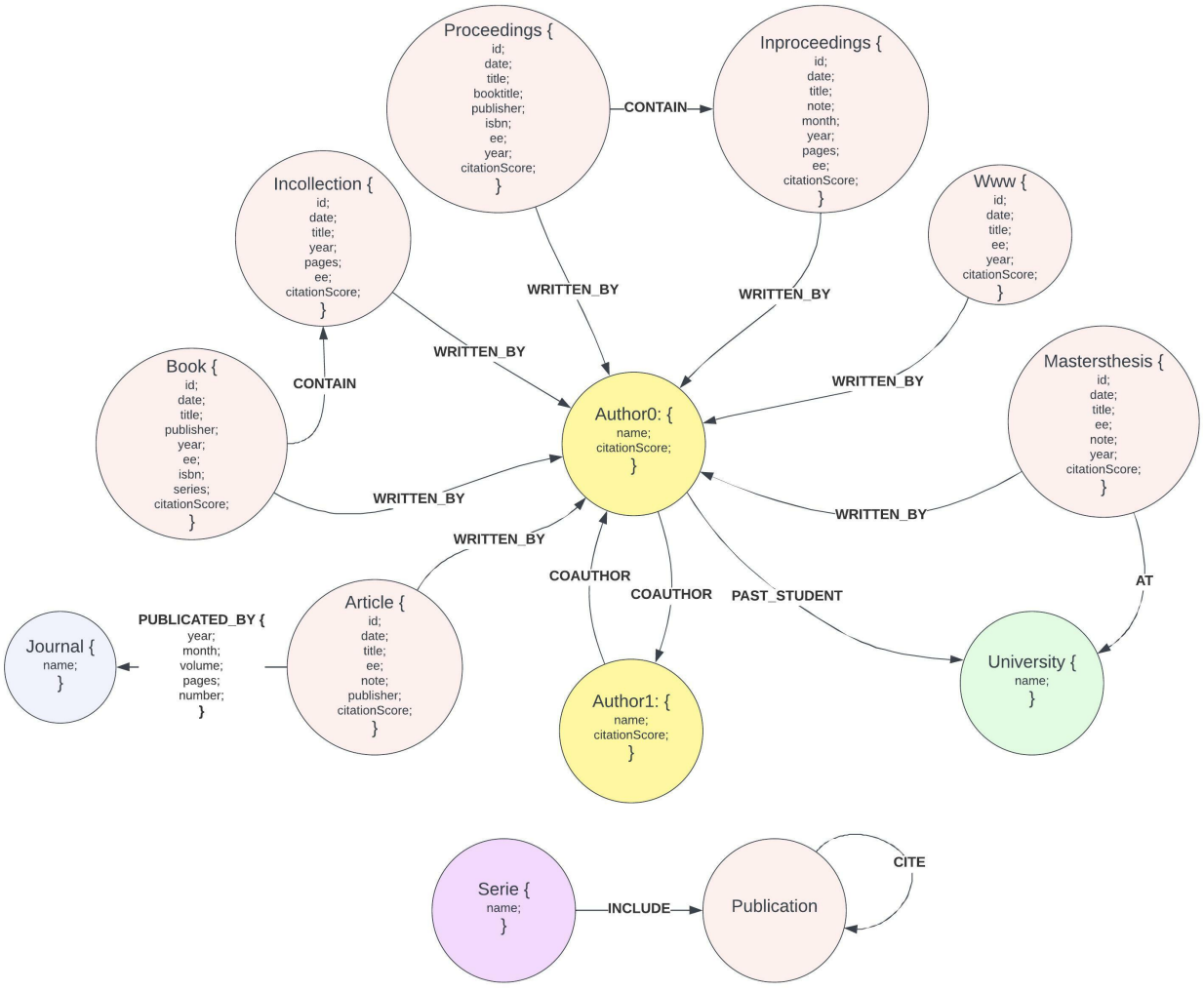



Figure 3.1: Graph Diagram

## 3.2. ER Diagram and Neo4j difference

The complete and general ER model that has been created represents an example of a real scientific bibliography. As requested in neo4j we worked in order to get the best from the tool used and the type of database. There are some differences in the treatment of the authors in neo4j (simple nodes with some attributes) and the authors in the ER diagram (researchers + generalization in a generic person) to differentiate the ones with ORCID and the ones without, since in reality only those who own the ORCID can write a book or scientific article, while students or other generic people can still speak in conferences (Proceedings), write online resources (Www) or write a thesis (Mastersthesis).Journal entity is more detailed in the ER diagram (with weak entities such as Volume and Number reflecting several more specific releases, and an entity “Publisher” that publish a book or a proceeding on the journal) then the graph diagram with some attributes in the relationship “Publicated_by”. The University node is not an entity in the ER diagram, but only a simple attribute, that’s because there are a few Universities connected only for thesis, the node is a way to create a circular relationship with the author (membership relationship to a school that is not explicit in the ER diagram to avoid redundancy). We have several authors in the graph diagram to have a "co-author" relationship that simply are authors of the same publications (or others). Finally, the difference in the treatment of the series entity which, thanks to the parent entity "Publication", is treated in a more orderly and concise way in the ER diagram compared to neo4j (due to the absence of true generalizations).

# 4 | Dataset Description and processing

The chosen Dataset is the DBLP computer science bibliography. It is the on-line reference for bibliographic information on major computer science publications. As January 2019 DBLP contains over 4.4 million publications, published by more than 2.2 million authors. The Dataset is divided in different types of publication (article, book, incollection, proceedings, inproceedings, www, mastersthesis) followed by their features and details. The attributes and relationships of these types are already described in the ER Diagram section 2. The Dataset has been subject to a process of elaboration and extraction in order to clear the data and take care of their quality, indeed, limited quality data could "cause a series of consequences for efficiency". The starting point was cut the 3.5 Gigabytes of data into 100 Megabytes (more or less 200.000 nodes) and convert its XML format into CSV. These operations were made much easier through the use of two Python programs:

1. The first one 4.1 aims at converting the entity names or entity numbers of the HTML characters with its original character representation the encoded UTF-8 string and at cleaning up the non-standard xml code. It is important to highlight that the program realizes the reading of the first 100 Megabytes of the Dataset performing a data cut.

```python
import html
import fileinput

totMB = 100

inputFile="D:/Uni/Magistrale - 1/SMBUD/Progetto/Data/dblp.xml"
outputFile="D:/Uni/Magistrale - 1/SMBUD/Progetto/Data/out"+str(totMB)+"M.xml"

input = open(inputFile,'rb')
text = input.read(totMB*1000000)
output = open("temp.xml",'wb')
output.write(text)
output.close()

output = open(outputFile, 'a', encoding="UTF-8")
output.truncate(0)
count = 0

for lines in fileinput.input(["temp.xml"]):
    line = lines.replace("<i>", "")
    line = line.replace("</i>", "")
    line = line.replace("<sup>", "")
    line = line.replace("</sup>", "")
    line = line.replace("<sub>", "")
    line = line.replace("</sub>", "")
    line = line.replace("&lt;", "")
    line = line.replace("&gt;", "")
    line = line.replace('&amp;', " and ")
    uLine = html.unescape(line)
    output.write(uLine)
    count+=1
    if count % 1000000 == 0:
        print(count)


output.close()
```

Figure 4.1: First Python program

2. The second one 4.2 carries out the conversion from XML to CSV and removes the special characters from CSV. One more thing to mention is that after all types of publications are found only a limited number is stored for each type (e.g. a maximum number of the type book and of the type article is saved fixing an upper bound at 2000).

```python
import xml.etree.ElementTree as ET
import csv
import math
import random
import copy

input_file = "out100M.xml"
num_limit = 2000

tree = ET.parse(input_file)
root = tree.getroot()

csvIndex = {}
csvHeaders = {}
keyListProceedings = []
keyListBooks = []

#find all types of pubblications and loads to a dictionary
for child in root:
    if child.tag not in csvIndex:
        csvIndex[child.tag]={}
        csvHeaders[child.tag]=[]

#scan all pub types
#maxDict object to store the maximum number of attributes for each pubblication type
for pub_type in csvIndex.keys():
    maxDict = {}
    currIndex = 2
    counter = 0

    for elem in root.findall(pub_type):
        thisDict = {}

        if pub_type == "proceedings":
            keyListProceedings.append(elem.attrib["key"])
            #save the key of that article in a list for randomic assigment

        if pub_type == "book":
            keyListBooks.append(elem.attrib["key"])
            #save the key of that article in a list for randomic assigment

        for innerTag in elem:
            #counts the number of each tags for an element
            if innerTag.tag in thisDict:
                thisDict[innerTag.tag]+=1
            else:
                thisDict[innerTag.tag]=1
```

```
48                #we create the tag in the dictionary
49
50            #updates the total count for that pubblication type
51            if innerTag.tag in maxDict:
52                if thisDict[innerTag.tag] > maxDict[innerTag.tag]:
53                    maxDict[innerTag.tag] = thisDict[innerTag.tag]
54            else:
55                maxDict[innerTag.tag]=1
56                #we create the tag in the dictionary
57
58        counter += 1
59        if counter>=num_limit:
60            break
61
62    #load the found number of attributes for
63    #that pubblication type in the global model of indexes
64    for key in maxDict.keys():
65        csvIndex[pub_type][key] = currIndex
66        currIndex = currIndex+maxDict[key]
67
68        csvHeaders[pub_type] = ["date","key"]
69        for key in maxDict.keys():
70            #creates the header for the csv
71            for n in range(maxDict[key]):
72                csvHeaders[pub_type].append(key+str(n))
73
74
75
76 #loads the data in the csv one type at time
77 for pub_type in csvIndex.keys():
78     counter = 0
79     out_csv = open("./csv/"+pub_type+'.csv','w', newline='', encoding='UTF-8')
80     csvwriter = csv.writer(out_csv)
81
82     csvwriter.writerow(csvHeaders[pub_type])
83     #write the header row
84
85     for elem in root.findall(pub_type):
86         row = [""] * len(csvHeaders[pub_type]) #array initialization
87         tmpIndex = copy.deepcopy(csvIndex)
88         #duplicate the index model for writing
89
90         #load fixed attributes of pubblications
91         row[0] = elem.attrib["mdate"]
92         row[1] = elem.attrib["key"]
93
94         #load dynamic values of tags
```

```
95          for innerTag in elem:
96              row_pos = tmpIndex[pub_type][innerTag.tag] #get the write column position
97              if innerTag.tag == "crossref" and pub_type == "incollection":
98                  row[row_pos] = random.choice(keyListBooks)
99              elif innerTag.tag == "crossref" and pub_type == "inproceedings":
100                 row[row_pos] = random.choice(keyListProceedings)
101             else:
102                 row[row_pos] = innerTag.text
103                 #write the corrisponding value
104
105             tmpIndex[pub_type][innerTag.tag]+=1
106             #increment the writing position
107
108         csvwriter.writerow(row)
109         counter += 1
110         if counter>=num_limit:
111             break
112
113     out_csv.close()
```

Figure 4.2: Second Python program

The final CSV result of this process is further analyzed and elaborated by deciding what to include or not in the Neo4j DB based on quantity of data present (erasing the weak and irrelevant data and relations). After doing this, the relationship between the field Book_Incollection and Proceeding_Inproceeding are generated randomly because the Database is not considered in its entirety, indeed, it is not certain that the original correlations have been remained unaltered or have been lost during the cutting operation. The last step has been the DB upload in Neo4j through the Cypher queries in particular using the command Load as we will see in section 5.18 .

# 5 | Queries and Commands

## 5.1. Query 1

This query finds for the specify journal X the authors that have written for it and the number of their respective publications in a certain period of time. The matched results are sorted in descendent through the ORDER BY clause and the query returns a limited subset of the rows through the LIMIT clause. In this example the query returns the top 10 author of IWBS Report during the period between 1990 to 2000.
Complexity 3 nodes + conditions + aggregations + limit

```
WITH '1990' AS startYear, '2000' AS endYear, 'IWBS Report' AS journalName
MATCH (aut:Author)-[:WRITTEN_BY]-(a:Article)-[r:PUBLICATED_BY] ->(j:Journal)
WHERE j.name = journalName AND r.year >= startYear AND r.year <= endYear
WITH aut, j, COUNT(*) AS numArtXAut
ORDER BY numArtXAut DESC
RETURN aut.name, numArtXAut
LIMIT 10
```

| | Name of the author | Number of articles x author |
|---|---|---|
| 1 | Christoph Beierle | 8 |
| 2 | Egon Börger | 7 |
| 3 | Stefan Böttcher | 6 |
| 4 | Ulrike Schwall | 5 |
| 5 | Stefan J. Schierholz | 4 |
| 6 | Udo Pletat | 4 |
| 7 | Gregor Meyer | 3 |
| 8 | Jochen Dörre | 3 |
| 9 | Ulrike Weiland | 3 |
| 10 | Claudia Maienborn | 3 |

## 5.2. Query 2

This query returns for each work of a specific author the number of entries(Incollection) that he has written in that Book, this values is computed through the ratio between the number of entries of the book written by the author and the total number of entries of the book. At the end we use the CASE clause to generate a conditional expression that returns the book's name and 0 if the author didn't write any entries of the book or the book doesn't have any entries. In this case the query returns the percentage of entries for each book written by Tor Helleseth.
Complexity 3 nodes + conditions + aggregations

```
MATCH (inc:Incollection)-[:WRITTEN_BY]->(aut:Author),(b:Book)-[:CONTAIN]->(inc)
WHERE aut.name = 'Tor Helleseth'
WITH b, COUNT(*) AS numIncXBook
MATCH (b)-[:CONTAIN]-(inc:Incollection)
WITH b, numIncXBook, COUNT(*) AS numTotIncXBook
RETURN b.title,
CASE
WHEN numIncXBook = 0 THEN 0.0
WHEN numTotIncXBook = 0 THEN 0.0
ELSE (numIncXBook * 1.0 / numTotIncXBook * 1.0) * 100.0 END AS Percentage
```

| | Book title | Rating |
|---|---|---|
| 1 | Handbook on Graph Drawing and Visualization | 5.26315 |
| 2 | Recommender Systems Handbook | 4.16667 |
| 3 | Encyclopedia of Complexity and Systems Science | 11.76471 |
| 4 | Handbook of Discrete and Computational Geometry, Second Edition | 12.5 |
| 5 | "Encyclopedia of Parallel Computing" | 5.26315 |
| 6 | Encyclopedia of Data Warehousing and Mining (4 Volumes) Second Edition (4 Volumes) | 4.54546 |
| 7 | Information security management handbook, Sixth Edition | 5.55556 |
| 8 | Encyclopedia of Computational Neuroscience | 5.0 |
| 9 | Handbook of Algorithms for Physical Design Automation | 4.34782 |
| 10 | Handbook of Formal Languages, Volume 2. Linear Modeling: Background and Application | 3.84615 |
| 11 | Handbook of Information and Communication Security | 4.54546 |
| 12 | Handbook of Semantic Web Technologies | 6.25 |
| 13 | Handbook of Mathematical Methods in Imaging | 6.66667 |
| 14 | Handbook of Virtual Environments - Design, Implementation, and Applications, Second Edition | 5.0 |
| 15 | Handbook of Spatial Logics | 5.26315 |
| 16 | Handbook of Database Security - Applications and Trends | 4.34782 |
| 17 | The Handbook of Mobile Middleware | 6.66667 |

## 5.3. Query 3

This query returns the article of journal with their respective author that are the published in a certain period of time. The result is sorted in descendent by the publication date and it is a limited subset of the rows through the limit clause. In this case the query returns in descendent order the 10 article of journal with their respective author during the period between 1990 to 2000.
Complexity 3 nodes + conditions + aggregations+limit

```
MATCH (aut:Author)<-[:WRITTEN_BY]-(a:Article)-[r:PUBLICATED_BY]->(j:Journal)
WHERE r.year>='1980' and r.year<= '1990'
WITH a.title AS articleTitle, r.year AS pubYear, aut.name as Autore
ORDER BY pubYear DESC
RETURN articleTitle,Autore, pubYear
LIMIT 10
```

| | articleTitle | author | pubblication year |
|---|---|---|---|
| 1 | Object Data Model Facilities for Multimedia Data Types | Mark F. Hornick | 1990 |
| 2 | Object Data Model Facilities for Multimedia Data Types | Frank Manola | 1990 |
| 3 | Object Data Model Facilities for Multimedia Data Types | Alejandro P. Buchmann | 1990 |
| 4 | Object Oriented DBMS as Generalization of Relational DBMS | David Beech | 1990 |
| 5 | Object Oriented DBMS as Generalization of Relational DBMS | Cetin Ozbutun | 1990 |
| 6 | Modification of User Defined Types | Phil Shaw | 1990 |
| 7 | Modelling Spatial Knowledge on a Linguistic Basis | Geoffrey Simmons | 1990 |
| 8 | Modelling Spatial Knowledge on a Linguistic Basis | Ewald Lang | 1990 |
| 9 | Modelling Spatial Knowledge on a Linguistic Basis | Kai-Uwe Carstensen | 1990 |
| 10 | Das Problem der VP-Ellipsen im Rahmen einer Theorie der Sachverhaltsanaphern | Elke Busch | 1990<br>1990 |

## 5.4. Query 4

This query returns the title of works published by a specific journal over the last year with the authors who wrote it. The max() function returns the maximum value in a set of values and after the matching condition are applied. In this case the query returns the works published in the last year by the journal IWBSReport and their authors
Complexity 3 nodes + conditions + aggregations + limit

```
MATCH(a:Author)<-[:WRITTEN_BY]-(p:Publication)-[r:PUBLICATED_BY]-(j:Journal)
WITH j,r,p,a,max(r.year) AS maxYear
WHERE j.name = 'IWBSReport' AND r.year = maxYear
RETURN p.title,collect(a.name) AS authors
LIMIT 10
```

| | p.title | authors |
|---|---|---|
| 1 | ILFA Interface Concept | ["Wolfgang Schönfeld", "Theodor Lettmann", "Bernd I. Dahn"] |
| 2 | Modelling Spatial Knowledge on a Linguistic Basis | ["Geoffrey Simmons", "Ewald Lang", "Kai-Uwe Carstensen"] |
| 3 | Domain Restriction and Other Kinds of Reference Set Operations in Sentence and Discourse Semantics. | ["Rolf Mayer"] |
| 4 | Das Problem der VP-Ellipsen im Rahmen einer Theorie der Sachverhaltsanaphern | ["Elke Busch"] ["Elke Busch"] |
| 5 | Syntaktische Informationseinheiten zu deutschen Adjektiven | ["Stefan J. Schierholz"] |
| 6 | Complexity of Logical Decision Problems: An Introduction | ["Egon Börger"] |
| 7 | Position und Bewegung: Zur Semantik lokaler Verben | ["Claudia Maienborn"] |
| 8 | On Subsumption and Semiunifaction in Feature Algebras | ["William C. Rounds", "Jochen Dörre"] |
| 9 | Derivationsmorphologie in einem Übersetzungssystem | ["Anne Schiller"] |
| 10 | Knowledge Representation in LILOG | ["Kai von Luck", "Udo Pletat"] |

## 5.5. Query 5

This query returns a ranking of the first 10 authors that wrote on a specific journal. The score of each author is calculated using the average citation score of its publications adding a weight based on the number of publications. After matching all publications of an author for that journal, loading them as a collection using the collect() function and elaborating them, the results are ordered based on the FinalScore value and limited to 10 using the LIMIT function.
Complexity 3 nodes + conditions + aggregations + limit

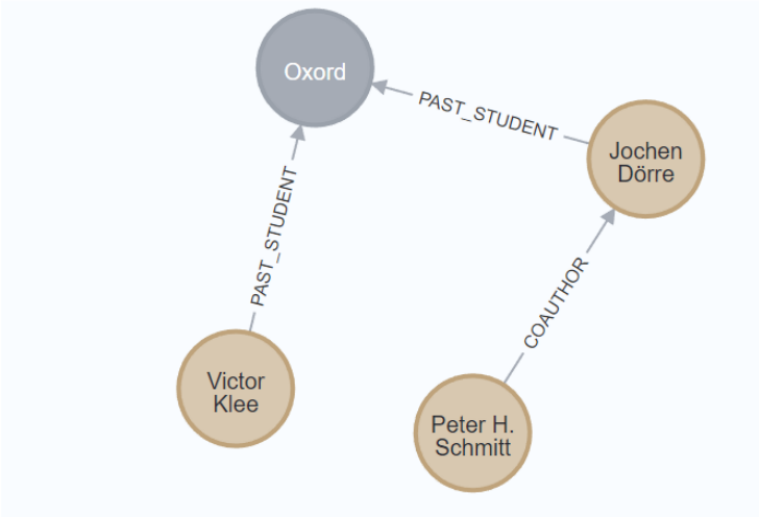
```
MATCH (a:Author)<-[:WRITTEN_BY]-(p:Publication)-[:PUBLICATED_BY]->(j:Journal)
WHERE j.name='Sci. Eng. Ethics'
WITH collect(p) AS pubs, sum(p.citationScore) AS
totScore,a, size(collect(p)) AS numPubs
RETURN a.name as Name, totScore, numPubs, totScore/numPubs AS
meanScore, 0.1*totScore+0.9*(totScore/numPubs) AS FinalScore
ORDER BY FinalScore DESC
LIMIT 10
```

| | Name | totScore | numPubs | meanScore | FinalScore |
|---|---|---|---|---|---|
| 1 | Michael D. Mumford | 16.00781 | 16 | 1.00048 | 2.50122 |
| 2 | Stephanie J. Bird | 14.94547 | 15 | 0.99636 | 2.39127 |
| 3 | Luciano Floridi | 15.08879 | 16 | 0.94304 | 2.35762 |
| 4 | Jaime A. Teixeira da Silva | 13.24581 | 14 | 0.94612 | 2.17609 |
| 5 | Neelke Doorn | 13.00399 | 14 | 0.92885 | 2.13637 |
| 6 | David B. Resnik | 12.07406 | 12 | 1.00617 | 2.11296 |
| 7 | Tanuj Kanchan | 11.86092 | 12 | 0.98841 | 2.07566 |
| 8 | Ana Marusic | 11.64288 | 12 | 0.97024 | 2.03750 |
| 9 | Shane Connelly | 11.63754 | 12 | 0.96979 | 2.03657 |
| 10 | Kewal Krishan | 10.98351 | 11 | 0.99850 | 1.99700 |

## 5.6. Query 6

This query uses the neo4j function shortestPath() to find the shortest path between two nodes. In this case we search for authors "PeterH.Schmitt" and "Victor Klee". To have a relevant result we impose a minimum of 2 nodes in the shortest path. This query could be used to find the fastest path to put in communication two authors who do not know each other.

```
MATCH(Peter:Author{name:"PeterH.Schmitt"}), (Victor:Author{name:"VictorKlee"}),
p=shortestPath((Peter)-[*]-(Victor))
WHERE length(p)>1
RETURN p
```




## 5.7. Query 7

This query returns a ranking of the first 10 authors that wrote on a journal in a given year. The score of each author is calculated using the average citation score of its publication adding a weight based on the number of publications. The result is ordered using the FinalScore of each author. In this case the used year for the query is 1997.

Complexity 3 nodes + condition + aggregations + limit

```
MATCH(a:Author)<-[:WRITTEN_BY]-(p:Publication)-[r:PUBLICATED_BY]-(j:Journal)
WHERE r.year='1997'
WITH collect(p) AS pubs,sum(p.citationScore) AS
totScore,a,size(collect(p)) AS numPubs
RETURN a.name AS Name,totScore,numPubs,totScore/numPubs AS
meanScore,0.1*totScore+0.9*(totScore/numPubs) AS FinalScore
ORDER BY FinalScore DESC
LIMIT 10
```

| | Name | totScore | numPubs | meanScore | FinalScore |
|---|---|---|---|---|---|
| 1 | Victor Klee | 1.35527 | 1 | 1.35527 | 1.35527 |
| 2 | Peter Gritzmann | 1.35527 | 1 | 1.35527 | 1.35527 |
| 3 | Jochen Bern | 1.30958 | 1 | 1.30958 | 1.30958 |
| 4 | Christoph Meinel | 4.67460 | 5 | 0.93492 | 1.30888 |
| 5 | Helmut Seidl | 2.92863 | 3 | 0.97621 | 1.17145 |
| 6 | Oleg V. Khamisov | 1.15070 | 1 | 1.15070 | 1.15070 |
| 7 | Christian Fecht | 2.02935 | 2 | 1.01467 | 1.11614 |
| 8 | Reiner Horst | 1.11138 | 1 | 1.11138 | 1.11138 |
| 9 | Le Dung Muu | 1.11138 | 1 | 1.11138 | 1.11138 |
| 10 | Belinda B. King | 1.07856 | 1 | 1.07856 | 1.07856 |

## 5.8. Query 8

This query returns the ranking of the first 4 universities ordered by the number of students who have completed a thesis at it. The number of thesis is found using the size() function. The result is ordered according to the number of past students at that university. Complexity 3 nodes + conditions + aggregations + limit

```
MATCH (p:Author)<-[:WRITTEN_BY]-(t:Mastersthesis)-[:AT]-(u:University)
MERGE(u)<-[:PAST_STUDENT]-(p)
WITH collect(t.title) AS Thesis, u, size(collect(t.title)) AS
numPublication, collect(p.name) AS
pastStudents,size(collect(p.name)) AS numPastStud
RETURN u.name AS Univiersity, Thesis, numPublication AS NumOfPublic,
pastStudents, numPastStud
ORDER BY numPastStud DESC
LIMIT 4
```

| | Univiersity | Thesis | NumOfPublic | pastStudents | numPastStud |
|---|---|---|---|---|---|
| 1 | Cambridge | ["Portierung des DBLP-Systems auf ein relationales Datenbanksystem und Evaluation der Performance.", Who's Who in the World Wide Web: Approaches to Name Disambiguation", "Der Einfluss kleiner naturnaher Retentionsmaßnahmen in der Fläche auf den Hochwasserabfluss -Kleinrückhaltebecken -."] | 3 | ["Stephan Vollmer", "Vanessa, C. Klaas", "Rita Ley"] | 3 |
| 2 | Stanford | ["PRPL: A Database Workload Specification Language, v1.3."] | 1 | ["Kurt P. Brown"] | 1 |
| 3 | Harvard | ["Efficient View Maintenance at Data Warehouses."] | 1 | ["Tolga Yurek"] | 1 |
| 4 | Oxord | ["Regelbasierte Extraktion und asymmetrische Fusion bibliographischer Informationen."] | 1 | ["Oliver Hoffmann 0002"] | 1 |

## 5.9. Query 9

This query take from the "MIT" University the best 10 students for the number of journals that have published their publications, considering only publications after 2000. The reverse arrow relationship search was also used to show the centrality of the Publication node in the query.
Complexity 3 nodes + conditions + aggregations + limit

```
MATCH(u:University)<-[:PAST_STUDENT]-(s:Author)<-[:WRITTEN_BY]-(p:Publication)-
[:PUBLICATED_BY]->(j:Journal)
WHERE p.date>="2000-01-01" AND u.name="MIT"
WITH s.name AS student,collect(DISTINCT(j.name))
AS journals,size(collect(j.name)) AS  numJournal
RETURN student,journals,numJournal
ORDERBY numJournal DESC
LIMIT 10
```

| | student | journals | numJournal |
|---|---|---|---|
| 1 | Dieter Baum | ["Universität Trier, Mathematik/Informatik, Forschungsbericht"] | 21 |
| 2 | Stephanie J. Bird | ["Sci. Eng. Ethics"] | 15 |
| 3 | Tanuj Kanchan | ["Sci. Eng. Ethics"] | 12 |
| 4 | Stefan Böttcher | ["IWBS Report"] | 9 |
| 5 | Michael W. Kalichman | ["Sci. Eng. Ethics"] | 8 |
| 6 | Bert Gordijn | ["Sci. Eng. Ethics"] | 8 |
| 7 | Aceil Al-Khatib | ["Sci. Eng. Ethics"] | 7 |
| 8 | Roland Seiffert | ["IWBS Report", "LILOG-Memo", "LILOG-Report"] | 6 |
| 9 | Jochen Bern | ["Universität Trier, Mathematik/Informatik, Forschungsbericht"] | 6 |
| 10 | Patricia Osseweijer | ["Sci. Eng. Ethics"] | 6 |

## 5.10. Query 10

In this query we search the ranking of Universities by number of citations in 2019, so we have to search for the relations with every author that was a student in the specific University and sum up their number of publications.
Complexity 3 nodes + conditions + aggregations + limit

```
MATCH (u:University)<-[:PAST_STUDENT]-
(a:Author)<-[:WRITTEN_BY]-(p1:Publication)<-[c:CITE]-(p2:Publication)
WHERE '2019-01-01'<=p1.date <= '2020-01-01'
WITH u.name AS University, size(collect(p2)) AS NumPublication
RETURN University, NumPublication
ORDER BY NumPublication DESC
LIMIT 5
```

| | University | NumPublication |
|---|---|---|
| 1 | Toronto University | 3219 |
| 2 | Cambridge | 1515 |
| 3 | MIT | 1470 |
| 4 | Oxord | 1462 |
| 5 | Polimi | 1058 |

## 5.11. Query 11

This query classifies the Universities that have the highest number of students that have published in the economic field (to do this it is used the Serie node), limited to five.
Complexity 3 nodes + conditions + aggregations + limit

```
MATCH(s:Serie)-[:CONTAIN]->(p:Publication)-[:WRITTEN_BY]->(a:Author)
[past:PAST_STUDENT]->(u:University)
WHERE s.name="AllAboutEconomic"
WITH collect(a.name) AS Students, size(collect(a)) AS Num_Students, u
RETURN Students, Num_Students, u.name
ORDER BY Num_Students DESC
LIMIT 5
```

| | Students | Num$_{S}$*tudents* | u.name |
|---|---|---|---|
| 1 | ["Jong Yong Abdiel Foo", "Marc Vauclair", "Brian Bowe", "Tim Berners-Lee", "Jong Yong Abdiel Foo", "Dirk Schnelle"] | 6 | Cambridge |
| 2 | ["Olivier Ridoux", "Anne Canteaut", "Diana Adela Martin", "Roy T. Fielding", "Paul C. Leyland"] | 5 | Oxord |
| 3 | ["Tor Helleseth", "Rosalie J. Ocker", "Eddie Conlon"] | 3 | Stanford |
| 4 | ["Henrik Frystyk Nielsen", "Helmut Seidl"] | 2 | Harvard |
| 5 | ["Friedrich L. Bauer"] | 1 | MIT |

## 5.12. Query 12

In this query there is a match of series that contains a type of publications with the quotes to other publications. After there is (again) a match with all the quotes to publications (that are in the specific serie, except other that are already in that serie). Finally there is a collect of the number of citations (found before) in a best 5.
Complexity 3 nodes + conditions + aggregations + limit

```
MATCH(s:Serie)-[:CONTAIN]->(p1:Publication)<-[c:CITE]-(p2:Publication)
WITH  s , collect(p2) AS pubs, p1
MATCH (p1)<-[c2:CITE]-()
WHERE  NOT p1  IN pubs
WITH size(collect(c2)) AS NumCitation2,s
RETURN s.name, NumCitation2
ORDER BY NumCitation2 DESC
LIMIT 5
```

| | s.name | NumCitation2 |
|---|---|---|
| 1 | AllAboutEconomic | 924 |
| 2 | AllABoutComputerScience | 642 |
| 3 | AllAboutAi | 471 |
| 4 | AllAboutSocialScience | 316 |
| 5 | AllAboutPhysic | 188 |

## 5.13. Query 13

This query classify the 5 most important authors by the number of coauthors they have. To work it uses the following functions from gds and apoc libraries:

- Gds.graph.drop (to drop a graph based on the data input)
- Gds.graph.project (to create a sort of view without any directed link)
- Apoc.when
- Gds.degree.stream (to use the "view" created)

```
CALL gds.graph.exists('coauthorsUn')
YIELD exists AS ex
CALL apoc.when(ex, 'CALL gds.graph.drop("coauthorsUn") YIELD graphName AS G' )
YIELD value AS val
```

```
CALL gds.graph.project( 'coauthorsUn', 'Author', {
COAUTHOR: { orientation: 'UNDIRECTED' }})
YIELD graphName AS coauthorsGraph
CALL gds.degree.stream(coauthorsGraph)
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).name AS authorName, score AS numbeOfCoauthor
ORDER BY numbeOfCoauthor DESCENDING, authorName
LIMIT 5
```

| | authorName | numbeOfCoauthor |
|---|---|---|
| 1 | Ana Marusic | 48.0 |
| 2 | Michael D. Mumford | 36.0 |
| 3 | Ivan Buljan | 35.0 |
| 4 | Ellen-Marie Forsberg | 29.0 |
| 5 | Hub A. E. Zwart | 26.0 |

## 5.14. Query 14

For a certain university named "MIT", this query returns the ranking of the 3 most famous proceedings counting the total number of citations (proceeding + inproceedings) from 2000 onwards.
Complexity 5 nodes + conditions + aggregations + limit

```
MATCH (u:University)<-[:PAST_STUDENT]-(aut:Author)<-[:WRITTEN_BY]-
(p:Proceedings)-[:CONTAIN]->(i:Inproceedings)<-[:CITE]-(pub:Publication)
WHERE u.name="MIT" AND pub.date >="2000-01-01"
WITH size(collect(pub)) AS NumCitation , p, aut
MATCH (p)<-[:CITE]-(pub2:Publication)
WITH aut, p, NumCitation+size(collect(pub2)) AS TotNumCitation
RETURN aut.name, p.title, TotNumCitation
ORDER BY TotNumCitation DESC
LIMIT 3
```

| | aut.name | p.title | TotNumCitation |
|---|---|---|---|
| 1 | Joaquim Filipe | Transactions on Computational Collective Intelligence XVII | 283 |
| 2 | Joaquim Filipe | Transactions on Computational Collective Intelligence XX | 195 |
| 3 | Stéphane Loiseau | Transactions on Computational Collective Intelligence XX | 195 |

## 5.15. Command 1

This command is used to update the citationScore attribute of the publications. In the first part of the command it is checked if already exist a graph called "citationsGraph" in the gds library. If this graph already exist, it is dropped. In the second part of the command we create a new graph in the gds library (using the command project) containing all the CITATION relationships between publications. Finally on this graph it is launched the PageRank algorithm (gds.pageRank.stream) which calculates all the indexes that will be written as attribute in the corresponding publication nodes.

```
CALL gds.graph.exists('citationsGraph')
YIELD exists AS ex
CALL apoc.when(ex, 'CALL gds.graph.drop("citationsGraph") YIELD graphName as G' )
YIELD value as val
CALL gds.graph.project( 'citationsGraph', 'Publication', 'CITE')
YIELD graphName AS graph
CALL gds.pageRank.stream(graph)
YIELD nodeId, score
MATCH(n:Publication)
WHERE id(n) = nodeId
SET n.citationScore =score
```

## 5.16. Command 2

This command deletes all publication nodes, proceeding and inproceeding linked to a certain author. In this case the name of the author is: "Dieter Baum" .

```
MATCH (aut:Author)<-[:WRITTEN_BY]-(proceedings:Proceedings)
WHERE aut.name ="Dieter Baum"
WITH proceedings AS p, aut
OPTIONAL MATCH (p:Proceedings)-[:CONTAIN]->(inproceedings:Inproceedings)
WITH p,aut, inproceedings AS inp
DETACH DELETE p, inp
```

## 5.17. Command 3

This command creates an article published in a journal and that cite other publications of a certain author. In this case the article created is WRITTEN_BY “Paul Kocher”, PUBLICATED_BY “meltdownattack.com” and CITE "Spectre Attacks: Exploiting Speculative Execution." and "An Evaluation of Object-Oriented DBMS Developments: 1994 Edition."
Complexity 3 nodes + conditions + aggregations

```
WITH ["Spectre Attacks: Exploiting Speculative Execution.",
"An Evaluation of Object-Oriented DBMS Developments: 1994 Edition."]
AS citatedList
MATCH(p:Publication)
WHERE p.title IN citatedList
WITH collect(p) AS allCitated,p
MERGE(article:Article:Publication{citationScore:0.0,date:date(),ee:"https://=)",
id:" tr/marhanz/s00",title:"Happy Birthday Algoritm"})-[:WRITTEN_BY]->
(aut:Author{name:'Paul Kocher'})
MERGE (article)-[:PUBLICATED_BY {year:date().year}]->
(j:Journal{name:'meltdownattack.com'})
MERGE (article)-[:CITE]->(p)
```

## 5.18. Extra Commands

This section introduces some of the necessary operation used to structure the db by creating Nodes and their relationships. These operations are essentials to perform the previous queries.

### 5.18.1. Import in the Database

As mentioned in the previous chapter, the DB is imported in Neo4j from csv file through the cypher queries in particular using the command Load. Below are shown a couple of example where Www, Authors and Editor are upload in the database with their attributes. The Merge clause ensures that a pattern exists in the graph when the specified pattern is not present and needs to be created, any variables previously bound to existing graph elements will be reused in the pattern. With Header clause we are specifying that the file starts with a single row containing column names allowing specific fields access by their corresponding column name.

#### Load Www Nodes

```
LOAD CSV WITH HEADERS FROM 'file:///www.csv' AS row
WITH row
MERGE (www:Www:Publication {id: row.key})
ON CREATE SET www.date = row['date'], www.title = row['title0'],
www.ee = row['ee0'], www.year = row['year0']
```

#### Load Authors of Www Nodes

```
LOAD CSV WITH HEADERS FROM 'file:///www.csv' AS row
WITH row
MATCH (www:Www {id: row.key})
UNWIND range(0,4) AS i
WITH i, www, row WHERE row['author'+toString(i)] IS NOT NULL
MERGE (aut:Author {name: row['author'+toString(i)]})
MERGE (www)-[:WRITTEN_BY]->(aut)
```

#### Load Www Editors as Authors

```
LOAD CSV WITH HEADERS FROM 'file:///www.csv' AS row
WITH row
MATCH (www:Www {id: row.key})
UNWIND range(0,5) AS i
WITH i, www, row WHERE row['editor'+toString(i)] IS NOT NULL
MERGE (aut:Author {name: row['editor'+toString(i)]})
MERGE (www)-[:WRITTEN_BY]->(aut)
```

### 5.18.2. Creation Coauthor relationship

This command creates a coauthor relationship between author of the same article and returns some data to analyze it. The list of author that has written the specified article is realized through the collect() function, then for each different authors the coauthor relationship is created, the query returns a limited subset of the rows through the limit clause.

```
MATCH (n:Article)-[w:WRITTEN_BY]->(a:Author)
WITH n,collect(a) AS authors
FOREACH( a1 IN authors |
FOREACH( a2 IN [a IN authors WHERE a<> a1| a]|
MERGE (a1)-[:COAUTHOR]-(a2)))
RETURN *
LIMIT 300
```

### 5.18.3. Creation Cite relationship

This command creates randomly Cite relationship between publication avoiding self-quotation. The range() returns a list comprising all integer values within a range bounded by a start value start and end value, in this case an interval included between 1 and 120 as citationRange is considered. The apoc.coll.randomItem(collection) function returns a random item from the list and it is used to randomly choose the number of cite that each publication has and which are the cite they are linked to,then for each different publication the Cite relationship is created.

```
WITH range(1,120) AS citationRange
MATCH(p:Publication)
WITH collect(p) AS publications, citationRange
MATCH(p2:Publication)
WITH p2, apoc.coll.randomItems(publications,
apoc.coll.randomItem(citatioRange))  AS citatedPublications
FOREACH (citated IN [c IN citatedPublications WHERE c <>p2 | c]  |
CREATE (p2)-[:CITE]->(citated))
```

```
MATCH (p:Author)<-[:WRITTEN_BY]-(t:Mastersthesis)-[:AT]-(u:University)
MERGE(u)<-[:PAST_STUDENT]-(p)
```